

Do High Schools Shape College Choices? Evidence from Centralized Assignment in Chile

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Abstract

School quality is usually measured with test scores, but high schools may also shape students' college choices. This paper studies this question in Chile, where centralized assignment creates lottery variation in high-school offers. I link assignment records to administrative data on school enrollment, standardized achievement, and higher-education applications and enrollment. I estimate observational school value added for math, verbal achievement, enrollment in high-premium institutions and high-premium fields, and projected program income, using a rich set of pre-treatment controls. I then test whether lottery-induced attendance at higher-value-added schools improves the corresponding outcome, controlling for expected value added in each student's assignment portfolio. Achievement forecast coefficient estimates are 0.884 for math and 0.885 for verbal achievement, close to benchmarks from settings with lagged test scores. Higher-education value added also predicts causal changes in later choices: pass-through is 0.497 for high-premium-institution enrollment, 0.803 for high-premium-field enrollment, and 0.311 for log projected program income. Heterogeneity estimates suggest that the pass-through of math value added varies across the baseline math achievement distribution. Gender also reveals important heterogeneity in achievement pass-through, with value added more informative for girls. Lastly, an ordered value-added horse race shows that projected-income gains are not primarily summarized by math value added, but rather by other dimensions of value added. High schools therefore shape college choices through dimensions of school value that achievement value added alone does not capture. Choices that appear to reveal student preferences may thus partly reflect the information, expectations, and preparation produced by the high school attended.

1 Introduction

School quality is usually summarized by achievement. This is natural: test scores are widely available, comparable across schools, and even predictive of later outcomes (Chetty et al. 2011). But school and teacher effects are not necessarily one-dimensional. Recent work shows that schools and teachers can affect attendance, behavior, crime, teenage motherhood, labor-market outcomes, and other long-run outcomes in ways that are only weakly related to test-score effects (Jackson 2018; Beuermann et al. 2023; Rose et al. 2022; Angrist et al. 2023). Families and policymakers might therefore care about school quality measures that capture more than just achievement.

This broader view of school quality is especially relevant for high schools. High schools are the last formal educational institutions students attend before making higher-education choices, and those choices are immensely consequential. Whether students enter higher education, which institutions they attend, and which fields they choose all matter because returns vary substantially across institutions and, especially, fields of study (Kirkeboen et al. 2016). Yet many measures of school effects on higher education are coarse, focusing on indicators for enrollment or persistence rather than on the type of institution or field students enter.

There are also good reasons to think that higher-education choices are shaped by frictions that schools may influence. Prior work documents large differences in application and enrollment behavior across groups of students, and shows that relatively small interventions can change college search, application behavior, and major choice (Hoxby and Turner 2015; Dynarski et al. 2021; Hastings et al. 2016; Chetty et al. 2025). If high schools differ in counseling, information provision, expectations, peer norms, academic confidence, or preparation for particular fields, then their effects on higher education may extend beyond their effects on achievement.

This possibility also changes how higher-education choices should be interpreted. Decisions not to continue to higher education, to avoid particular fields, or to attend lower-quality institutions are often viewed as revealing stable preferences or privately optimal responses to expected returns. Yet expected returns and the options students perceive may themselves depend on the information, preparation, and support provided by their high school. A choice can therefore be privately optimal given beliefs and preparation that the school helped produce, while still being causally affected by the school attended. This distinction is especially relevant in settings where families have limited information about educational options and access to counseling varies across schools.

This paper asks whether high schools causally affect higher-education choices, and whether those effects are captured by traditional achievement value added. The main empirical challenge arises from the fact that students and families sort across schools. Families choose schools based on resources, preferences, networks, location, and expectations; these same characteristics may also shape later higher-education choices. As a result, differences across schools may reflect both causal school effects and the underlying characteristics of the students and families who select into them.

I address this challenge using Chile’s centralized school assignment system, the Sistema de Admisión Escolar (SAE). SAE assigns students to public and private-subsidized schools through a centralized mechanism that uses random tie-breakers when schools are oversubscribed. The system covers a large national school sector, allowing me to study school effects at a scale that is unusual in lottery-based designs. This assignment variation can be linked to administrative data covering school enrollment, standardized tests, and higher-education applications and enrollment. Linking random assignment in the mechanism to higher-education choices allows me to answer this question in a national context.

The analysis proceeds in two steps. First, I estimate observational value added for each school and outcome in the broad panel. These measures are school fixed effects from regressions that adjust for rich pre-treatment controls, including grade-4 standardized test scores, family-background variables, and middle-school controls. Second, I validate these value-added measures for each school using the SAE lottery offers as exogenous variation. I instrument for attended school with the school offered through the centralized mechanism, while controlling for the expected value added that arises from the probabilities of assignment of the student to the schools in their list, together with the value added measured in the first step.

I summarize this relationship with a forecast coefficient, ψ , in line with Angrist et al. (2017)¹. If $\psi = 1$, then the causal effect induced by complying with a lottery offer equals the gain predicted by observational value added for that outcome. If $\psi = 0$, the observational value-added measure has no causal predictive content along the lottery-induced margin. The main estimates suggest meaningful pass-through. For achievement, the estimates are 0.884 for math and 0.885 for verbal achievement. For higher-education choices, the forecast coefficients are 0.497 for enrollment in a high-premium institution, 0.803 for enrollment in a high-premium field, and 0.311 for log projected program income. In practical terms, schools that observationally look better at raising achievement or higher-education outcomes also predict causal effects on those same outcomes, though the causal effects are smaller than the value-added might suggest.

The paper contributes to three literatures. First, it contributes to work on teacher and school value added. A large literature estimates the effects of teachers and schools on test scores and longer-run outcomes (Chetty et al. 2014; Dobbie and Fryer 2015; Jackson 2018; Beuermann et al. 2023; Rose et al. 2022). This work increasingly emphasizes that test-score value added may miss important dimensions of educational production. I extend this idea to high-school effects on higher-education choices, focusing not only on enrollment or achievement but also on whether students enter high-premium fields, high-premium institutions, and programs with higher projected earnings.

Second, the paper contributes to the literature using lotteries and centralized assignment systems to estimate school effects (Abdulkadiroğlu et al. 2014; Abdulkadiroğlu et al. 2017; Angrist et al. 2024). In the spirit of Angrist et al. (2017), I ask whether observational value-added measures are useful guides to causal school effects. The contribution here

¹I refer to the forecast coefficient as pass-through interchangeably

is to apply this pass-through logic to a national high-school assignment system and to outcomes observed after students transition into higher education.

Third, the paper contributes to work on higher-education choice. Prior research shows that the field and institution students choose matter for later outcomes, and that information and application frictions can change those choices (Kirkeboen et al. 2016; Hoxby and Turner 2015; Dynarski et al. 2021; Hastings et al. 2016; Chetty et al. 2025). I study high schools as a possible upstream source of these differences. Rather than evaluating a particular counseling or information intervention, the paper measures whether the high-school system itself generates causal differences in students' higher-education paths.

The rest of the paper is organized as follows. Section 2 describes the Chilean setting and the linked administrative data. Section 3 presents the empirical framework. Section 4 reports the main results, heterogeneity by baseline achievement and gender, and the ordered value-added horse race for projected program income. Section 5 concludes.

2 Context and Data

2.1 Institutional Setting

Chile is a useful setting for this project because national administrative data records school enrollment, academic histories, standardized achievement, and higher-education choices with individual identifiers that allow students to be matched across educational data.

Formal secondary education begins in grade 9 and lasts four years. Students may attend public, private-subsidized, or private non-subsidized schools. Public and private-subsidized schools are fully or partly financed by the State and thus participate in the centralized school assignment system described below. They serve 90.6% of the high-school population. Private non-subsidized schools remain outside that mechanism and serve the remaining 9.4%.

2.2 Centralized School Assignment

Chile’s centralized school assignment system, the Sistema de Admisión Escolar (SAE), assigns students to participating schools through a version of student-proposing Deferred Acceptance. Families submit ranked lists of schools. The mechanism combines these rankings with school vacancies and priority rules, and it uses random tie-breakers when there are more applicants than seats within a priority group. If a student’s offered school depends on one of these tie-breakers in the system, different lottery draws will imply different school assignments. We say therefore this student is at assignment risk. Conditional on the information that determines priorities and assignment probabilities, these tie-breakers or lotteries generate the quasi-experimental variation in school assignment for students with assignment risk as described by Abdulkadiroğlu et al. (2017).

SAE started in 2017 in some regions and was rolled out nationally in 2018. Given that I study high school effects, I use applications to grade-9 vacancies between 2018 and 2021. These cohorts are expected to finish high school between 2023 and 2025, allowing us to observe high school attendance and post-schooling outcomes.

Implementing this design requires recovering each applicant’s probability of assignment to each school among those in the submitted rankings of schools. I do this by replicating the SAE assignment algorithm and repeatedly simulating counterfactual lottery draws, which determines which students are at assignment risk. As a diagnostic check, when I run the replicated algorithm using the original lottery numbers for the 2018–2020 regular grade-9 processes, I reproduce over 99% of observed first-round assignments in each of the three cohorts.² The remaining mismatch reflects institutional features that are difficult to fully recover from public documentation, but the high replication rate gives confidence that the simulated probabilities capture the relevant assignment risk.

²Even though SAE runs two additional rounds, these complementary rounds are very limited in sample size. Since they assign vacancies in previously undersubscribed schools, assignment risk is unlikely in these rounds.

This centralized system is very relevant to determine the high school attended. Figure 1 makes this point by showing attendance rates in the 100 most demanded schools, with different markers for applicants with and without first round offers.

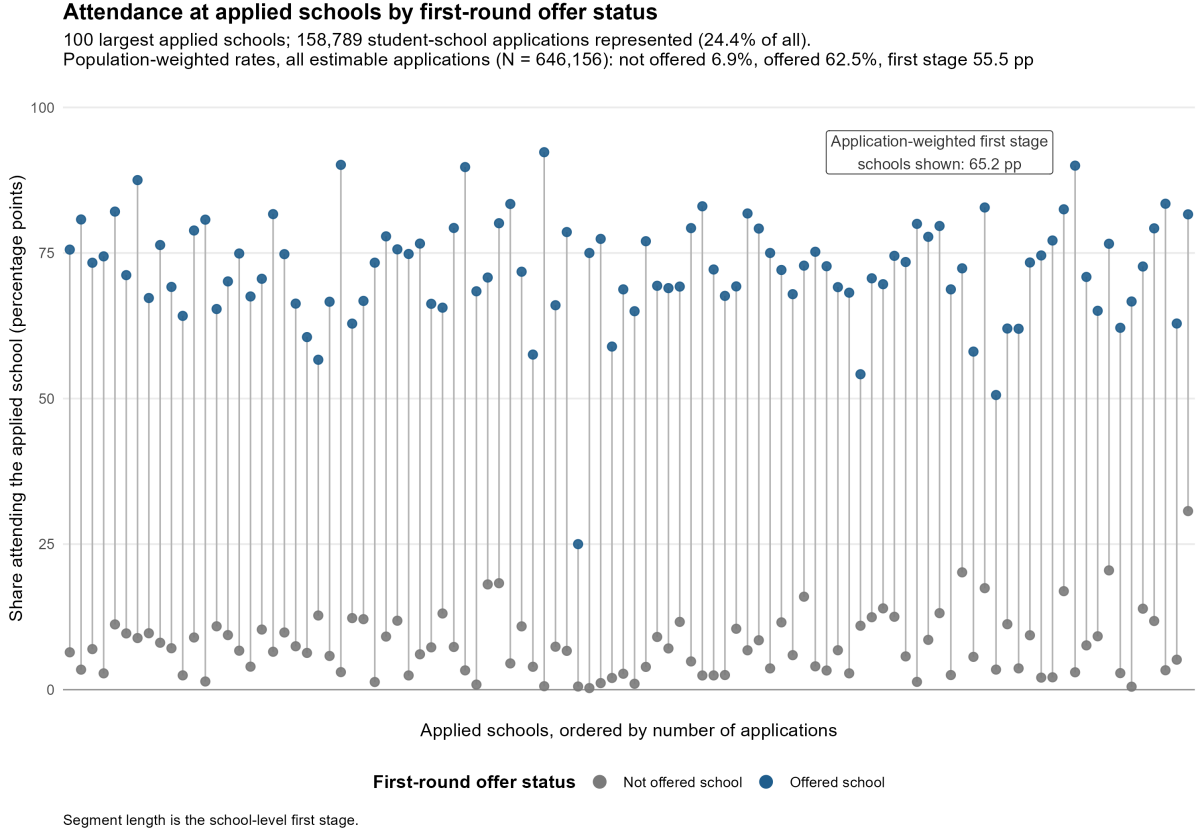


Figure 1: Effect of SAE offers on attendance

For this exercise and the rest of the paper, I define high-school attendance as the school where a student spends the most time after grade 8, with most recent spells breaking ties. This definition smooths over short enrollment spells and transfers, and it assigns a single high-school value to each student. As a result, students who did not receive a first-round offer might still end up attending a certain high-school via complementary rounds or an application cycle in a later year. On average, a high school is attended at a rate of 6.9% by the first-round rejected students and at a rate of 62.5% by the accepted students. This suggests an average effect of the first-round offer of 55.5 percentage points, making it a relevant instrument for school attendance.

2.3 Sample Description

In this subsection I describe the samples I employ for the two steps of my analysis. In my first step, I construct measures of value added of schools in different outcomes as residuals in a regression with rich covariates and as representative as possible. I refer to the sample used for that exercise as the value added sample. In the second step, I use the exogenous variation in the centralized mechanism to study whether school changes

due to tie-breakers are as predictive of effects as the value added metric suggest. For that exercise I collect all students with assignment risk in their grade-9 SAE application. I refer to this sample as the lottery sample.

2.3.1 Value Added Sample

I construct the value added sample by merging annual enrollment, grade progression, attendance, grades, and completion records using the student identifier provided for educational data. The linked administrative panel starts from the universe of students first observed in grade 8 between 2017 and 2020, producing 944,510 students across four cohorts, defined by their year of grade-8 enrollment. For instance, a student in the 2019 cohort who wants to change schools within the non-private system applies to SAE that year, is expected to start high school in 2020, and is expected to graduate in 2023.

To construct value-added for each school I keep as much from this broad panel as possible, while ensuring the comparison is reasonable. From this universe I drop students younger than 12 or older than 16 in grade 8 and those who do not attend high school.

Given that value-added estimation relies on a rich set of baseline covariates, this becomes the most important restriction on this sample. For these covariates, I use information from Chile’s national standardized assessment system (SIMCE) and middle school relative performance. Middle school controls are constructed by comparing attendance and GPA of a student to those of their peers in middle school every year. These are then averaged and summarized into two middle school z-scores for a student: one for attendance and one for GPA.

SIMCE exams are administered in different grades depending on the year. This always includes grade 4 and occasionally grades 8 and/or 10. Unfortunately grade-8 and grade-10 measures are not available for all the studied cohorts. I therefore use grade-4 SIMCE as the main pre-treatment achievement control.

SIMCE also collects parent surveys. I use these surveys to measure family background before treatment, including household income, parental education, indigenous origin of the parents, and early-childhood education. These variables are useful because they proxy for family resources and prior investments that may influence both high-school choices and later higher-education outcomes. When parent-survey variables are missing, I use middle school and neighborhood cells to impute medians where enough donor observations are available; remaining missing values are left missing.

As can be seen in Panel A of Table 1, from the original 944,510 students in these cohorts, I keep the 757,999 for which I have a complete control vector with all the sources of information described above. Sample restrictions defined in each row should be read cumulatively within each panel.

Table 1: Construction of the value-added and SAE samples

Panel A: Value-added sample		Panel B: SAE lottery sample	
Sample restriction	Students	Sample restriction	Students
Cohorts 2017–2020	944,510	Cohorts 2018–2020	706,042
+ valid HS and age 12–16	931,547	+ valid HS and age 12–16	696,817
+ complete VA control set	757,999	+ complete lottery control set	587,142
+ admission test	622,809	+ application in SAE	241,490
		+ assignment risk	135,916
		+ admission test	108,418

2.3.2 Lottery Sample

For the lottery sample I start following virtually the same steps but with the 2018–2020 cohorts, the first ones with SAE rolled out at a national level. Panel B of Table 1 repeats the accounting exercise above for the lottery sample. From an original list of 706,402 we can track the baseline set of controls for only 587,142. Out of that group only about 41% (241,190) applied for a change of school through the SAE mechanism for a spot in grade 9³. In some cases, students (e.g., those applying to undersubscribed schools) are guaranteed an offer to a certain school. The assignment mechanism does not yield exogenous variation for such cases since there is no tie-breaking. As a result, the SAE lottery sample (135,916) only considers those students whose preferences imply non-trivial assignment probabilities.

The broad complete-control panel is used to estimate the main school value-added measures, while the SAE lottery exam-taker sample is used to validate score-based value-added measures using assignment variation. For some outcomes, I condition the samples to test-takers which is when the last row becomes relevant.

2.4 Characterization of the samples

How different are these two samples? Table 2 helps us answer this question. Here, I report means on some key variables for the broad panel (column 1), students that applied for a HS in SAE (column 2) and their subset which had applications with assignment risk (column 4).

³I consider only applications to grade 9 in the year the student is in grade 8, to avoid including students who changed schools after failing a grade

Table 2: Balance across analysis samples

	Broad Panel (All HS)	SAE Applicants		SAE Applicants w/ risk		Risk SAE - All SAE
	Mean	Mean	Diff. (2)-(1) (SE)	Mean	Diff. (4)-(1) (SE)	Diff. (4)-(2) (SE)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Student characteristics</i>						
Female	0.492	0.484	-0.008 (0.001)	0.495	0.004 (0.001)	0.011 (0.001)
Metropolitan Region (Santiago)	0.387	0.257	-0.130 (0.001)	0.273	-0.114 (0.001)	0.016 (0.001)
Age	13.4	13.4	0.000 (0.001)	13.3	-0.035 (0.001)	-0.035 (0.001)
<i>Baseline SIMCE variables</i>						
Math 4th grade (z)	0.036	-0.232	-0.268 (0.002)	-0.159	-0.195 (0.002)	0.073 (0.002)
Verbal 4th grade (z)	0.036	-0.189	-0.225 (0.002)	-0.115	-0.151 (0.002)	0.074 (0.002)
Income decile	5.5	4.5	-0.991 (0.005)	4.8	-0.721 (0.006)	0.271 (0.004)
Mother education years	11.9	10.9	-1.002 (0.005)	11.2	-0.709 (0.007)	0.292 (0.005)
% matched all SIMCE vars	82.4%	81.8%		83.4%		
<i>Middle school</i>						
GPA (z)	0.032	0.024	-0.008 (0.001)	0.093	0.061 (0.002)	0.069 (0.001)
Attendance (z)	0.021	0.030	0.009 (0.001)	0.043	0.022 (0.002)	0.014 (0.001)
% matched all middle-school vars	99.8%	99.8%		99.8%		
<i>Outcomes</i>						
Std. math exam	0.000	-0.334	-0.334 (0.002)	-0.289	-0.289 (0.002)	0.045 (0.001)
Std. verbal exam	0.000	-0.301	-0.301 (0.002)	-0.230	-0.230 (0.003)	0.072 (0.002)
Admission exam taker	0.776	0.738	-0.038 (0.001)	0.769	-0.007 (0.001)	0.031 (0.001)
Higher ed. enrolled	0.672	0.580	-0.092 (0.001)	0.617	-0.055 (0.001)	0.037 (0.001)
Hi-premium institution	0.128	0.056	-0.072 (0.000)	0.067	-0.061 (0.001)	0.011 (0.000)
Hi-premium field	0.307	0.258	-0.048 (0.001)	0.280	-0.026 (0.001)	0.022 (0.001)
<i>N</i>	931,547	294,198		162,633		
Percent of broad panel	100.0%	31.6%		17.5%		
Joint test <i>p</i> -value			< 0.001		< 0.001	< 0.001

Notes: Broad Panel comprises cohorts 2017–2020 with an observed attended high-school RBD and age 12–16. SAE Applicants are timely grade-9 applicants in cohorts 2018–2020; *w/ risk* further requires $\max_j p_{ij} < 1$ in the full simulated assignment distribution. Difference columns report raw mean differences: columns (2)–(1), (4)–(1), and (4)–(2), respectively. Parentheses contain influence-function standard errors that account for overlap between the groups. The joint test covers the baseline characteristics above the Outcomes section. Percentages are relative to the Broad Panel. The SIMCE and middle-school blocks report complete-variable match rates.

Column (3) report differences between SAE applicants and the broad sample. From there we can see that the sample of students that apply for a new school in grade 9 are negatively selected relative to the population: For instance, they perform around 0.2 SD below in achievement tests in grade 4, on average have mothers with one fewer year of education and perform 0.3 SD worse in exams to higher education. The joint test clearly rejects that the two populations are similar.

Column (5) shows a comparable pattern, but with smaller differences. Column (6) makes it clear that students with assignment risk are positively selected relative to those without, with the respective joint test also rejecting that the two groups are similar.

The differences present in (5) could make one worry that estimates run on the broad panel might not be useful to predict effects on the population of SAE applicants with assignment risk. For this to be an issue, SAE applicants at-risk would need to be different from those of the broad sample in ways that our rich set of controls does not capture. If

that were the case, the validation step of the design would show that the VA estimated are biased and not as predictive of causal effects. At its core, the validation step answers whether estimates in sample (1) predict causal effects in sample (4).

2.5 Higher Education Outcomes

The higher-education system also generates administrative data on applications, admission scores, and enrollment. Taking the national admission exam is a choice for secondary-school students, but it is the standard route into selective higher education. Students who apply through the centralized admission system take mandatory math and verbal components, and may also take optional subject exams.⁴ I use the mandatory math and verbal scores as achievement outcomes, standardizing them within test year to make them comparable across test regimes.

Higher-education applications and admissions are organized at the program-institution level. A student's admission depends on entrance-exam scores, school GPA-based scores, ranked program preferences, and the weights each program assigns to score components. This structure is useful because it makes higher-education choices much richer than a single college-enrollment indicator: students choose not only whether to enroll, but also which institution, program, and field to pursue.

Students in the sample have not yet finished higher education, so I cannot observe their realized earnings. I take advantage of MiFuturo, an information source run by the Ministry of Education that reports labor-market statistics at the program level. I use its reported average income in the fourth year after graduation to assign an earnings proxy to higher-education options.

Because the MiFuturo income file is not observed at the exact enrollment-program code used in the matricula records, I do not treat it as a direct one-to-one match for every student-program record. Instead, I estimate log-income fixed-effect models in the MiFuturo data using institution and generic-program cells. Figure 2 shows the estimated generic-program fixed effects organized by field. I then use generic-program and institution estimates to construct projected-income measures for the matriculated programs observed in the student panel when the exact program has no direct MiFuturo value.

I link the school panel to higher-education enrollment records for the first observed enrollment. I use three higher-education choice outcomes. The first is an indicator for enrollment in a high-premium field. This indicator equals one for students whose first observed higher-education enrollment is in Science, Law, Engineering/Manufacturing/-Construction, or Medicine+⁵, which correspond to the right-hand side of Figure 2.

The second is an indicator for enrollment in a high-premium institution. This indicator equals one when the student's first observed matriculated institution has a particularly

⁴The national admission exam was called the PSU through 2020, the PTU in 2021–2022, and the PAES from 2023 onward.

⁵Medicine+ includes premium health-related careers such as medicine, pharmacy, nursing, midwifery, medical technology, and dentistry. Other health and technical-health programs are not included in this high-premium-field category.

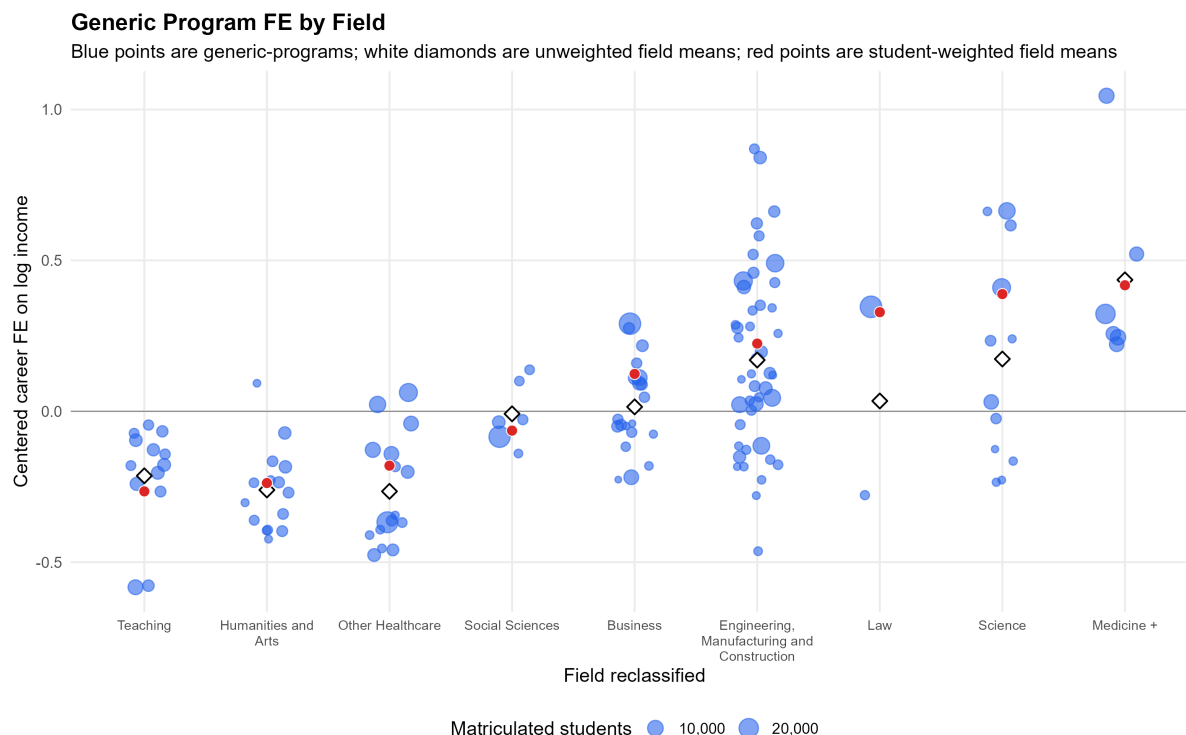


Figure 2: MiFuturo Generic-Program Income Premia by Field

high institution fixed effect in the MiFuturo log-income model.⁶ Students without an observed higher-education enrollment are coded as zero for the high-premium-field and high-premium-institution indicators.

The third is log projected program income. For matriculated students, projected program income comes from MiFuturo-based predictions using a hierarchy that first uses institution-by-generic-program fixed effects when available, then generic-program fixed effects, then institution fixed effects, and finally the global MiFuturo mean. For students with no observed higher-education enrollment, projected program income is set to the current minimum-wage proxy of 553,553 Chilean pesos before taking logs.⁷

[FIX] Talk about becas/ credits ?

⁶Specifically, I use a threshold of 0.1 log points relative to the average program, which roughly translates to an average institution premium of 10%.

⁷This is consistent with the fact that the amounts reported in MiFuturo are in 2025 Chilean pesos.

3 Empirical Framework

This section describes the empirical object and the research design. The goal is not to estimate a separate causal effect for every school and every possible margin of substitution. Instead, I ask whether an outcome-specific observational value-added measure is informative about causal effects generated by lottery-induced changes in school attendance. The design therefore collapses school effects into scalar measures and estimates a forecast coefficient for each outcome.

3.1 School Effects and Observational Value Added

I use a constant-effects value-added model as a guide:

$$Y_i^m(j) = \mu^m + V_j^m + \epsilon_i^m, \quad (1)$$

Here, $Y_i^m(j)$ is student i 's potential outcome for outcome m if assigned to high school j . The index m denotes the outcome, and j denotes the high school. The term μ^m is the average level of outcome m under the student-weighted average school. The term V_j^m is the contribution of school j to outcome m relative to that average. The term ϵ_i^m captures student-level determinants of outcome m that do not vary with the assigned school in this constant-effects representation. I normalize school effects so that their student-weighted mean is zero, $\sum_j \omega_j V_j^m = 0$, where ω_j is the share of students linked to school j in the value-added sample. This centering means that the weighted average of V_j^m across schools is exactly zero. A positive V_j^m therefore denotes a school above the student-weighted average for outcome m , while a negative value denotes a school below that average. The main outcomes are math and verbal achievement, indicators for enrollment in a high-premium field and a high-premium institution, and log projected program income.

The first step constructs observational value added in the broad complete-control panel. For each outcome m , I estimate

$$Y_i^m = \alpha^m + \lambda_{s(i)}^m + X_i' \Gamma^m + u_i^m, \quad (2)$$

where $s(i)$ is the student's main high-school RBD and X_i contains pre-treatment controls. These include cohort, gender, age, a third-degree polynomial in each of grade-4 SIMCE math and language, family-background measures from SIMCE parent surveys⁸, and middle-school controls from grades 5 to 8 (middle school indicator and school-normalized attendance and performance). I define the observational value-added measure $\hat{V}_j^m$ as the centered empirical analogue of V_j^m : the estimated school fixed effect $\hat{\lambda}_j^m$ from equation (2), centered using the same student-weighting convention.

Importantly, these fixed effects by themselves answer an observational question: conditional on X_i , which schools have higher average outcomes? If school assignment in the

⁸In particular: family income decile, years of education and indigenous status of each parent, and indicators for whether the student attended any institution before primary education.

broad panel were truly random, then the centered fixed effect $\hat{\lambda}_j^m$ would identify V_j^m up to sampling error. In practice, students sort across schools, so the fixed effects may combine true school effects with any remaining selection not absorbed by controls. The second step therefore asks whether the part of $\hat{V}_j^m$ shifted by the SAE lotteries corresponds to causal effects in the same outcome.

3.2 Assignment Risk and Expected Value Added

SAE applicants submit rank-ordered lists of schools. Let Z_i denote the first-round school offer and D_i denote the school attended, measured by the post-grade-8 most-time RBD. A student has a preference and priority type θ_i , and the assignment mechanism implies an assignment-probability vector $p_i = (p_{i1}, \dots, p_{iJ})$, where $p_{ij} = \Pr(Z_i = j)$. Under equal treatment of equals, students with the same type have the same assignment probabilities (Abdulkadiroğlu et al. 2017). For pre-treatment characteristics and potential outcomes fixed under re-randomization, we have

$$(Y_i^m(1), \dots, Y_i^m(J), X_i) \perp\!\!\!\perp Z_i \mid p_i. \quad (3)$$

The raw vector p_i is high-dimensional, because students can rank many different combinations of schools. For an outcome-specific scalar value-added measure, I summarize assignment risk by the expected value added in the student's portfolio:

$$E_i^m = \sum_j p_{ij} \hat{V}_j^m. \quad (4)$$

This is the average value added the student would receive over repeated randomizations of the assignment mechanism, holding the submitted application and priorities fixed. In the main IV specification, this scalar expected value replaces a large set of school-specific probability controls.

For each outcome m , define the value added of the attended school and offered school as

$$A_i^m = \hat{V}_{D_i}^m, \quad O_i^m = \hat{V}_{Z_i}^m. \quad (5)$$

The instrument is O_i^m , the value added of the school generated by the lottery offer. The endogenous variable is A_i^m , the value added of the school actually attended. The notation is mnemonic: O_i^m is the value added of the **O**ffered school, and A_i^m is the value added of the **A**ttended school. Because assignment offers vary randomly conditional on the assignment probabilities, variation in O_i^m around E_i^m provides quasi-experimental variation in the quality of the school offer.

3.3 Scalar IV Specification

The first empirical question is whether observational value added predicts causal effects on the same outcome. Specifically, when the assignment mechanism moves a student toward a school with higher $\hat{V}_j^m$, does outcome m rise, and by how much relative to the

observational difference encoded in $\widehat{V}_j^m$? The main IV system is given by the following pair of equations:

$$Y_i^m = \alpha^m + \psi^m A_i^m + \rho^m E_i^m + W_i' \delta^m + \eta_i^m, \quad (6)$$

$$A_i^m = a^m + \kappa^m O_i^m + \rho_A^m E_i^m + W_i' \delta_A^m + \nu_i^m. \quad (7)$$

Here, Y_i^m is student i 's realized value of outcome m . The value added of the attended school, A_i^m , is instrumented with the value added of the school offered through the assignment mechanism, O_i^m . E_i^m is the expected value added implied by the student's assignment-probability vector. Finally, W_i is a parsimonious lottery-sample control vector containing cohort, gender, age, and grade-4 SIMCE math and language. This differs from the richer control vector used to construct $\widehat{V}_j^m$ in equation (2), which also includes family-background and middle-school controls. All reported specifications use heteroskedasticity-robust standard errors.

The coefficient ψ^m is the key object. It measures how much of the observational value-added difference induced by the lottery offer passes through into causal effects on the same outcome. If $\psi^m = 1$, then a one-unit increase in observational value added causes a one-unit increase in the outcome. If $\psi^m = 0$, the observational measure has no causal predictive content along the lottery-induced margin. Values between zero and one imply that observational value added is directionally informative, but overstates the causal effect of attending higher-value-added schools.

The scalar design is useful because a fully flexible school-specific IV design would be high-dimensional and would mix many margins of substitution across fallback schools (Angrist et al. 2023; Kirkeboen et al. 2016). It instead asks whether movement toward schools with higher measured value added causes better outcomes, after controlling for the expected value added implied by each student's assignment risk.

A second empirical question is whether the forecast differs by students' baseline achievement. To answer that question, I also estimate equation (6) separately by quintiles of baseline grade-4 math achievement or gender. Because gender and grade-4 SIMCE are fixed before high-school assignment, these splits describe heterogeneity without conditioning on a post-treatment outcome.

3.4 Empirical Bayes

The school fixed effects in equation (2) are estimated with different precision. In particular, effects for schools with smaller regression samples or noisier outcomes contain more sampling error. I therefore use an Empirical-Bayes (EB) version of the observational school-value index, which shrinks noisier estimates more strongly toward the across-school mean (Walters 2024). For each outcome m , let se_j^m denote the regression-derived standard error of the student-weighted centered fixed effect $\widehat{V}_j^m$. This is the uncertainty of the same centered effect used in the scalar design, rather than the standard error of an arbitrary reference-normalized school dummy.

I estimate the latent across-school variance by subtracting average sampling variance from the student-weighted variance of the estimated effects:

$$\hat{\tau}_m^2 = \max \left\{ \text{Var}_\omega \left(\hat{V}_j^m \right) - \sum_j \omega_j (se_j^m)^2, 0 \right\}, \quad (8)$$

where ω_j is proportional to the number of students in school j 's stage-1 regression sample. The implied school-specific reliability is

$$\lambda_j^m = \frac{\hat{\tau}_m^2}{\hat{\tau}_m^2 + (se_j^m)^2}, \quad (9)$$

and the EB school value is

$$\hat{V}_j^{m,EB} = \bar{V}_\omega^m + \lambda_j^m \left(\hat{V}_j^m - \bar{V}_\omega^m \right). \quad (10)$$

Note that as $(se_j^m) \rightarrow 0$ (more precision on the estimate), $\lambda_j^m \rightarrow 1$ and the posterior approaches the original $\hat{V}_j^m$. Otherwise, the posterior lies between the estimate $\hat{V}_j^m$ and the weighted mean of such estimates, $\bar{V}_\omega^m$.

I recenter these posterior means using the same student weights, so their student-weighted mean remains zero.

I then replace $\hat{V}_j^m$ with $\hat{V}_j^{m,EB}$ everywhere in the scalar IV construction:

$$A_i^{m,EB} = \hat{V}_{D_i}^{m,EB}, \quad O_i^{m,EB} = \hat{V}_{Z_i}^{m,EB}, \quad E_i^{m,EB} = \sum_j p_{ij} \hat{V}_j^{m,EB}. \quad (11)$$

The EB specification instruments $A_i^{m,EB}$ with $O_i^{m,EB}$ and controls for $E_i^{m,EB}$ in the same way as equation (6).

3.5 Variance–Covariance Analysis

Equation (8) treats each outcome separately: it purges sampling noise from the variance of a single outcome's school-value estimates. But the paper's motivating claim, that school effects are not necessarily one-dimensional, is fundamentally a claim about how effects on *different* outcomes relate to one another across schools (Jackson 2018; Rose et al. 2022). Answering that question requires the covariance of the latent effects V_j^m and $V_j^{m'}$ across schools, not just their individual variances.

A naive covariance of the raw estimates, $\widehat{\text{Cov}}(\hat{V}_j^m, \hat{V}_j^{m'})$, is not a valid estimate of $\text{Cov}(V_j^m, V_j^{m'})$ for the same reason that the naive variance in equation (8) required a bias correction. Sampling error in $\hat{V}_j^m$ and sampling error in $\hat{V}_j^{m'}$ are both computed from the same students in the same regression sample, so they are generally correlated with each other: idiosyncratic cohort-level shocks, mechanical overlap in the estimation sample, or correlated measurement error can all generate a positive or negative covariance between the two sampling errors even if the underlying school effects V_j^m and $V_j^{m'}$ are unrelated. A plug-in covariance therefore conflates true cross-outcome covariation in school effects

with correlated noise, in a way that ordinary Empirical-Bayes shrinkage does not correct, since equation (8) only ever shrinks one outcome at a time.

To purge this bias, I use the cross-cohort estimator developed in Rose et al. (2022), adapting their teacher-by-year design to schools observed across grade-8 cohorts. The idea is to replace the single pooled estimate $\hat{V}_j^m$ with T_j separate, cohort-specific estimates for school j , and then use products of estimates from *different* cohorts to recover $V_j^m V_j^{m'}$ without the correlated-noise contamination that a single pooled cross-product would carry.

Formally, let $t \in \{1, \dots, T_j\}$ index the grade-8 cohorts observed for school j (cohorts 2017–2020), and define the school-by-cohort mean of the fitted residual plus school effect from equation (2):

$$\hat{V}_{jt}^m = \frac{1}{n_{jt}^m} \sum_{i: s(i)=j, c(i)=t} (\hat{\lambda}_j^m + \hat{u}_i^m) = V_j^m + \bar{u}_{jt}^m, \quad (12)$$

where $c(i)$ denotes student i 's grade-8 cohort and $\bar{u}_{jt}^m$ is the school-by-cohort sampling error, with $E[\bar{u}_{jt}^m] = 0$. Averaging equation (12) across a school's T_j cohorts recovers the pooled estimate: $\frac{1}{T_j} \sum_t \hat{V}_{jt}^m = \hat{V}_j^m$.

Assumption 1 (Uncorrelated cross-cohort sampling error). *For any school j and any two cohorts $t \neq t'$, and for any pair of outcomes m, m' (including $m = m'$), $E[\bar{u}_{jt}^m \bar{u}_{jt'}^{m'}] = 0$.*

Assumption 1 allows sampling errors in the same cohort to be correlated across outcomes – which is the problematic case above – but rules out shocks that persist or run across different grade-8 cohorts of the same school. It is the natural analogue, extended across outcomes, of the within-outcome independence assumption already implicit in the Empirical-Bayes correction in equation (8).

Under Assumption 1, $E_{t \neq t'} [\hat{V}_{jt}^m \hat{V}_{jt'}^{m'}] = V_j^m V_j^{m'}$, since all cross terms involving $\bar{u}_{jt}^m$ or $\bar{u}_{jt'}^{m'}$ vanish in expectation. Averaging this cross-cohort product over all $\binom{T_j}{2}$ distinct cohort pairs for school j , and aggregating across the J schools with $T_j \geq 2$, gives debiased estimators of the variance and covariance of the latent school effects:

$$\widehat{\text{Var}}(V^m) = \left(\frac{J-1}{J}\right) \frac{1}{J} \sum_j \binom{T_j}{2}^{-1} \sum_{t < t'} \hat{V}_{jt}^m \hat{V}_{jt'}^m - \frac{2}{J^2} \sum_{j < k} \hat{V}_j^m \hat{V}_k^m, \quad (13)$$

$$\widehat{\text{Cov}}(V^m, V^{m'}) = \left(\frac{J-1}{J}\right) \frac{1}{J} \sum_j \binom{T_j}{2}^{-1} \sum_{t \neq t'} \hat{V}_{jt}^m \hat{V}_{jt'}^{m'} - \frac{1}{J^2} \sum_{j \neq k} \hat{V}_j^m \hat{V}_k^{m'}. \quad (14)$$

Unlike equation (8), equations (13) and (14) require no explicit correction term built from regression standard errors. They are also not truncated at zero: they are unbiased for the latent variance and covariance directly, and can take small negative values in a finite sample even though the underlying quantity is, for the variance, non-negative by construction.

From equations (13) and (14), I construct two summary objects. The first is the implied correlation between latent school effects on outcomes m and m' ,

$$\hat{\rho}^{m,m'} = \frac{\widehat{\text{Cov}}(V^m, V^{m'})}{\sqrt{\widehat{\text{Var}}(V^m) \widehat{\text{Var}}(V^{m'})}}, \quad (15)$$

computed on the set of schools common to both outcomes' variance estimates. The second is the implied gain: the population-projection coefficient of $V_j^{m'}$ on V_j^m across schools,

$$\hat{\beta}^{m'|m} = \frac{\widehat{\text{Cov}}(V^m, V^{m'})}{\widehat{\text{Var}}(V^m)}. \quad (16)$$

$\hat{\beta}^{m'|m}$ answers a concrete question: if a school's true value added on outcome m were one unit higher, how much higher would its true value added on outcome m' be expected to be? Because $\widehat{\text{Var}}(V^m)$ and $\widehat{\text{Var}}(V^{m'})$ generally differ, $\hat{\beta}^{m'|m} \neq \hat{\beta}^{m|m'}$ in general.

I compute equations (13)–(16) for each pair among the same seven primary outcomes reported in Table 5: math and verbal achievement, admission-exam taking, high-premium-institution and high-premium-field enrollment, log projected program income, and financial-aid applications. If $\hat{\rho}^{m,m'}$ is close to zero, or $\hat{\beta}^{m'|m}$ is close to zero, then a school's true effect on outcome m carries little information about its true effect on outcome m' : the two dimensions of school quality are largely distinct, even after removing the noise that would otherwise attenuate their observed relationship. This is a purely observational counterpart to the value-added horse race below: it asks whether achievement value added and higher-education-choice value added are two names for the same underlying school quality, or whether they capture separate margins along which schools differ.

Standard errors for equations (13)–(16) are computed with a nonparametric bootstrap that resamples schools with replacement.⁹ For the correlation and implied-gain objects specifically, the resampled draw is joint: the same resampled set of schools is used to recompute $\widehat{\text{Var}}(V^m)$, $\widehat{\text{Var}}(V^{m'})$, and $\widehat{\text{Cov}}(V^m, V^{m'})$ together in each replicate, since these are ratios of quantities estimated from the same schools.

⁹Rose et al. (2022) instead derive analytic standard errors for these objects from the estimator's U-statistic representation. I use the bootstrap here as a first pass; incorporating the corresponding analytic formulas is left for a later draft.

4 Causal Effects of Schools

I organize the results in five blocks. First, I describe the scale of the underlying outcomes and the distribution of the seven primary observational school value-added measures. Second, I report the main forecast coefficient estimates for each outcome. Third, I examine heterogeneity in forecast by baseline grade-4 math achievement. Fourth, I examine whether pass-through differs by student gender. Finally, I use an ordered value-added horse race to compare achievement, high-premium-institution, and high-premium-field dimensions of school value added when predicting projected program income.

4.1 Distribution of Value-Added Estimates

Table 3 reports the scale of each student outcome and the dispersion of the corresponding school value-added measure for all seven primary outcomes. Binary outcomes and their value-added measures are expressed in percentage points, achievement outcomes in standard deviations, and projected income in log points. For school value added, the table compares the posterior EB standard deviation and 90–10 gap with the dispersion of the unshrunk estimates and the cross-cohort RSS estimate of latent dispersion.

Table 3: Scale of student outcomes and dispersion of school value added

Outcome	Units	Student outcome			School VA dispersion			
		<i>N</i>	Mean	SD	EB SD	Unshrunk SD	RSS SD	EB 90–10
Math achievement	SD	561,977	0.043	1.002	0.223	0.256	0.256	0.496
Verbal achievement	SD	569,047	0.049	0.989	0.151	0.187	0.187	0.375
Admission exam taken	pp	757,999	75.3	43.1	10.347	11.537	11.472	25.415
High-premium institution	pp	757,999	14.2	34.9	12.146	13.447	13.449	29.342
High-premium field	pp	715,740	34.7	47.6	5.579	8.448	8.562	12.561
Log projected income	log pts.	757,999	13.893	0.522	0.055	0.086	0.085	0.136
Financial aid application	pp	931,547	75.7	42.9	7.771	9.988	9.962	16.746

Notes: Each row uses the units shown in column 2: admission-test standard deviations (SD) for math and verbal achievement, percentage points (pp) for binary outcomes, and log points for projected income. The first block reports moments of the student outcome in the outcome-specific sample. The EB block reports the student-weighted standard deviation and 90–10 percentile gap of EB-shrunk, student-centered school value added. Schools are weighted by their number of students in the corresponding pooled value-added regression. Unshrunk VA is the student-weighted standard deviation of the school fixed-effect estimates supplied to the EB shrinkage step. RSS latent VA is the standard deviation of latent school effects estimated from cross-cohort products using the same sample rule and weights. All VA columns use the unified R-based pooled VA pipeline.

The school value-added measures also display substantial dispersion. The posterior EB standard deviation is 0.223 for math and 0.151 for verbal achievement; their 90–10 gaps are 0.496 and 0.375 standard deviations, respectively. For the binary outcomes, the posterior EB standard deviations range from 5.6 percentage points for high-premium-field enrollment to 12.1 percentage points for high-premium-institution enrollment. Projected-income value added has a posterior EB standard deviation of 0.055 log points and a 90–10 gap of 0.136 log points. As expected, EB shrinkage produces less dispersion than either the unshrunk estimates or the RSS estimates of latent school heterogeneity. These distributions show

that the observational measures contain economically meaningful variation and provide the scale needed to interpret the pass-through estimates below.

4.2 Validation of Value Added Estimates

In this section I report the results of the scalar IV regression.

4.2.1 First-stage estimates

Table 4 reports the first-stage coefficients corresponding to equation 7. In particular it reports the κ coefficient for each value added. This parameter summarizes how the value-added of the offers translates into improvements in the value-added of the school attended. For instance, students who receive an offer of a school with 1 unit s.d. higher VA in math achievement end up, on average, attending high schools with 0.442 more s.d. in math VA. Most coefficients are between 0.405 and 0.474, except for the first-stage coefficient on the value added for high-premium institutions which is estimated to be 0.269. This could suggest that this dimension of value added is either less salient or less relevant to families when they decide whether to comply or not with the offer. However, it could also be the case that this value added dimension is correlated with other elements which reduce the compliance.

Table 4: First-stage coefficients for the main EB school-value specifications

	Exams			Higher ed. choices			Financial aid
	Math	Verbal	Exam taking	High-premium inst.	High-premium field	Program income	Fin. aid app.
κ^{EB}	0.442*** (0.008)	0.435*** (0.007)	0.424*** (0.006)	0.269*** (0.009)	0.474*** (0.007)	0.439*** (0.006)	0.405*** (0.006)
First-stage F	3189.8	4210.0	5920.0	821.1	4278.6	4712.6	4222.7
N	92,361	93,980	135,844	94,487	94,487	94,487	135,844

Notes: Each column reports the first-stage coefficient κ^{EB} from regressing attended-school EB value added for the indicated outcome on first-round offered-school EB value added. The regressions and samples are exactly those underlying the corresponding columns of the main scalar-IV table. All specifications control for the DA-probability expected value of the same EB measure, cohort, grade-4 SIMCE math and language, gender, and age. Heteroskedasticity-robust standard errors are reported in parentheses. The reported F statistic tests the excluded offered-school value-added instrument. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

The first-stage F statistic is high across specifications confirming that the value added offer instrument is a relevant instrument for the value added of the school attended.

4.2.2 Second-stage estimates

Table 5 reports the seven primary EB IV estimates. Each column uses the same outcome to construct the observational school value-added measure, the first-round offered-school value-added instrument, and the realized attended-school value-added treatment. The coefficients can therefore be read as pass-through estimates: how much a lottery-induced increase in a school's observational value added translates into a causal gain in the corresponding student outcome.

Table 5: Pass-through coefficients for the main EB school-value specifications

	Exams			Higher ed. choices			Financial aid
	Math	Verbal	Exam taking	High-premium inst.	High-premium field	Program income	Fin. aid app.
ψ^{EB}	1.197*** (0.097)	1.031*** (0.098)	0.486*** (0.052)	0.667*** (0.120)	0.861*** (0.087)	1.060*** (0.131)	0.584*** (0.098)
p -value: $\psi^{EB} = 0$	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
p -value: $\psi^{EB} = 1$	0.042	0.754	< 0.001	0.005	0.108	0.649	< 0.001
N	92,361	93,980	135,844	94,487	94,487	94,487	135,844

Notes: School value added is estimated using grade-8 cohorts 2017–2020; the lottery sample uses SAE cohorts 2018–2020. Each column reports a scalar IV estimate of the pass-through from attended-school EB value added to the corresponding student outcome. Attended-school EB value added is instrumented with first-round offered-school EB value added. All specifications control for the DA-probability expected value of the same EB measure, cohort, grade-4 SIMCE math and language, gender, and age. Heteroskedasticity-robust standard errors are reported in parentheses. The two reported p -values test $\psi^{EB} = 0$ and $\psi^{EB} = 1$ using the same robust standard error. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The value-added metrics for achievement contain important signal: I reject $\psi = 0$ in both these dimensions of value added. In particular, the pass-through estimates for achievement are both above one: a one-unit increase in the EB math value-added index raises math achievement by 1.197, while the analogous estimate for verbal is 1.031. These coefficients are in the ballpark of those reported by Angrist et al. (2017) (in the range of 0.864 to 0.95) and those reported by Angrist et al. (2024) (in the range of 0.933 and 1.12 for conventional value added). There is evidence to reject the unbiased forecast for the math value added ($p = 0.042$) but I am unable to reject it for verbal value added ($p = 0.754$).

An important contribution of this paper is to extend this validation exercise to consequential higher-education outcomes for which a lagged measure of the same outcome is unavailable. The estimates show that the observational value-added measures contain causal predictive content well beyond achievement: the null hypothesis $\psi = 0$ is rejected for admission-exam taking, high-premium-institution enrollment, high-premium-field enrollment, projected program income, and financial-aid applications.

The magnitude of pass-through nevertheless differs across outcomes. A one-unit increase in admission-exam-taking value added raises the probability of taking the exam by 0.486 units, and the corresponding coefficient for high-premium-institution enrollment is 0.667. Both estimates are statistically below one, with $p < 0.001$ and $p = 0.005$, respectively, implying that the observational measures overpredict the gains generated by lottery-induced movement toward higher-value-added schools along these dimensions. Financial-aid-application value added also has a positive but incomplete pass-through of 0.584, and the one-for-one benchmark is rejected with $p < 0.001$.

By contrast, the pass-through coefficients for high-premium-field enrollment and projected program income are 0.861 and 1.060. For these outcomes, I cannot reject the one-for-one benchmark, with $p = 0.108$ for high-premium-field enrollment and $p = 0.649$ for projected program income. The estimates therefore suggest that observational differences across schools in these two dimensions closely predict the causal gains induced by the lottery, while the measures for exam taking, high-premium institutions, and financial-aid

applications remain informative but systematically overstate those gains.

Taken together, these results show that high schools affect more than test-score production. Lottery-induced movement toward schools with higher outcome-specific value added changes students' progression into the admissions process, the types of programs and institutions they enter, the projected earnings associated with those programs, and their engagement with the financial-aid system. Because these coefficients are identified by students whose attended-school value added responds to the lottery offer, they describe pass-through along the lottery-induced compliance.

4.3 Correlates of Value Added

The previous results show that schools differ along several distinct dimensions of value added.

My following step is trying to understand how value added varies across school groups and characteristics. The lottery design does not answer this question directly: lotteries shift which school a student attends, not the characteristics or the inputs that school uses. There is no exogenous variation in these results and thus, everything presented here is only descriptive – that is, correlations between value added and school characteristics. While this could hint at potential mechanisms it would be impossible to distinguish between the effect of, say, a low student-teacher ratio and the characteristics of schools willing and able to implement such ratio.

Table 6 shows a regression of each of the seven dimensions of value added studied on administration type, high-school size, and curricular tracks. Given that all value added measures are recentered at the population average, the coefficients on the intercept are meaningful. They show that medium-size municipal schools are estimated to have lower value added than the average school by 0.054 SD in mathematics and 0.027 SD in the verbal exam. These estimates suggest their students, by moving to an average school, would improve the rates they enroll in high-premium institutions and fields by 3.89 and 0.65 percentage points, respectively. They would also enroll in programs predicted to have 0.013 higher log points in predicted income.

In order to run these regressions I combine the school directory of individuals working on each school, their role and their level of education (check this). To get more specific, I merge this information with all the educational titles officially granted in Chile between 20XX and 20XX. This includes information all the undergraduate degree, master's degrees (3-4 semesters) as well as ABC (2 semesters degrees) and DEF (1 semester degrees). From this detailed information I measure which type of institutions staff got their degrees, as well as the name of the degree which I use to measure whether the staff has role-specific qualifications.

Because this detailed information is only available for degrees granted in the last 2X years, this level of detail is not available in the same way for all staff. Younger staff members would have their whole educational career tracked, but older staff would only have their most recent studies measured with detail.

Table 6: School value added by administration type, school size, and curricular track

	<i>N</i> schools	Share of students	(1) Math VA (SD)	(2) Verbal VA (SD)	(3) HP inst. VA (pp)	(4) HP field VA (pp)	(5) Income VA (log pts.)	(6) Exam VA (pp)	(7) Fin. aid app. VA (pp)
Intercept (omitted: Municipal, 100-249 students, no tracks)	84	1.3%	-0.054*** (0.007)	-0.027*** (0.005)	-3.887*** (0.389)	-0.650*** (0.244)	-0.013*** (0.002)	-0.081 (0.587)	0.076 (0.412)
<i>Administration type</i>									
Municipal corporation	222	8.8%	0.020* (0.011)	0.007 (0.009)	5.950*** (0.620)	0.331 (0.419)	0.012*** (0.004)	-2.209*** (0.810)	-2.106*** (0.575)
Private subsidized	1,795	52.8%	0.024*** (0.006)	0.021*** (0.005)	1.256*** (0.350)	0.193 (0.248)	0.010*** (0.002)	1.812*** (0.533)	0.837** (0.368)
Fully private	476	10.2%	0.330*** (0.014)	0.197** (0.008)	16.434*** (0.728)	1.739*** (0.298)	0.058*** (0.003)	6.523*** (0.632)	-10.119*** (0.603)
Delegated administration	70	4.4%	-0.021** (0.010)	-0.027*** (0.009)	1.358 (0.901)	2.822** (1.423)	0.009 (0.008)	-5.871*** (1.046)	-2.304*** (0.625)
SLEP	176	5.3%	0.012 (0.009)	0.013 (0.008)	3.137*** (0.815)	1.050** (0.416)	0.015*** (0.004)	-1.448 (0.928)	-0.949 (0.609)
<i>School size</i>									
Fewer than 100 students	626	3.6%	-0.042*** (0.009)	-0.033*** (0.006)	-1.674*** (0.408)	-0.262 (0.172)	-0.008*** (0.002)	-10.103*** (0.529)	-4.908*** (0.397)
250-499 students	977	36.8%	0.026*** (0.007)	0.021*** (0.005)	0.755* (0.408)	-0.155 (0.201)	0.000 (0.002)	2.748*** (0.392)	1.208*** (0.306)
500+ students	466	38.8%	0.022*** (0.008)	0.015** (0.006)	1.306** (0.547)	0.742** (0.343)	0.003 (0.003)	2.794*** (0.524)	1.530*** (0.379)
<i>School tracks</i>									
Technical-professional	931	40.8%	-0.101*** (0.005)	-0.111*** (0.004)	-1.309*** (0.337)	-0.051 (0.239)	-0.015*** (0.002)	-8.652*** (0.464)	-1.113*** (0.311)
Artistic	2	< 0.1%	-0.158*** (0.026)	0.024*** (0.007)	24.475*** (8.875)	-7.257* (3.922)	0.022 (0.036)	2.801 (1.800)	3.202*** (0.510)
Schools	3,274	100.0%							
Population mean			0.043	0.049	14.192	34.743	13.893	75.344	75.717
<i>R</i> ²			0.389	0.411	0.273	0.023	0.158	0.281	0.263

Notes: Each column is a school-level OLS regression of the indicated Empirical-Bayes value-added measure on administration-type, school-size, and curricular-track indicators. Outcomes are reported in their original units: math and verbal achievement in student standard deviations, binary outcomes in percentage points (pp), and projected income in log points. The omitted categories are Municipal DAEM, 100–249 estimated high-school students, and neither technical-professional nor artistic; the first row reports the intercept. All other rows report regression coefficients. There is no minimum-school-size restriction, and fully private schools are included. School size is estimated total enrollment across grades 9–12 using the four grade-8 cohorts. The first two columns report the number of schools and share of pooled value-added-sample students satisfying each row definition. Unconditionally, Municipal schools account for 18.4% of students and schools with 100–249 students account for 20.8%; these shares do not condition on the other omitted categories. The population-mean row reports the outcome-specific student mean around which each value-added measure is centered. The R^2 row reports the explanatory power of the full displayed specification. Size and track categories overlap administration types. Heteroskedasticity-robust (HC1) standard errors are in parentheses and do not incorporate estimation error in school value added. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

As a result, I construct different measures based on the age of the staff. (maybe use raw information for older and detailed for younger).

(Maybe have a first regresion which just has the more crude controls, and hwo different private schools might be).

(Also separate big and small schools?)

Table 7 reports school-level regressions of each Empirical-Bayes value-added measure on these characteristics. To help with interpretation both value added and the continuous regressors are standardized, so coefficients are in standard deviations of value added per standard deviation of the regressor.

All regressions include the administration-type, school-size-category, and curricular-track indicators reported in Table 6. These controls are suppressed from Table 7; geography is not included.

I limit this analysis to the 2,058 public and private-subsidized schools with at least 100 students in the value-added sample.

I exclude smaller schools to avoid extreme staffing ratios from particularly small schools.

The bottom rows of Table 7 summarize how much of each dimension these charac-

Table 7: Staff measurement coverage and school value added

	(1) Math VA	(2) Verbal VA	(3) HP inst. VA	(4) HP field VA	(5) Income VA	(6) Exam VA	(7) Fin. aid app. VA
<i>Staffing by age</i>							
Teachers: under age 35 per 100 students	0.171*** (0.033)	0.150*** (0.027)	-0.051** (0.021)	0.030 (0.027)	-0.018 (0.027)	0.107*** (0.020)	0.120*** (0.026)
Teachers: ages 35–49 per 100 students	0.096*** (0.030)	0.092*** (0.026)	0.014 (0.022)	0.027 (0.026)	0.029 (0.025)	0.035* (0.020)	0.031 (0.025)
Teachers: age 50+ per 100 students	0.034 (0.027)	0.035 (0.023)	0.023 (0.023)	0.077*** (0.030)	0.003 (0.027)	-0.047** (0.022)	-0.085*** (0.026)
Counselors: under age 35 per 100 students	-0.032 (0.028)	-0.045** (0.022)	-0.012 (0.016)	0.030 (0.025)	-0.008 (0.024)	-0.005 (0.017)	-0.056*** (0.019)
Counselors: ages 35–49 per 100 students	0.040 (0.025)	0.023 (0.023)	0.070*** (0.021)	0.038 (0.024)	0.043* (0.024)	0.019 (0.016)	0.033 (0.022)
Counselors: age 50+ per 100 students	0.014 (0.023)	0.005 (0.023)	0.030 (0.023)	-0.010 (0.031)	0.029 (0.031)	0.015 (0.017)	-0.013 (0.026)
Leadership: under age 35 per 100 students	0.021 (0.030)	0.012 (0.024)	-0.002 (0.025)	-0.034 (0.025)	0.009 (0.028)	-0.019 (0.020)	0.046* (0.026)
Leadership: ages 35–49 per 100 students	0.016 (0.025)	-0.011 (0.020)	0.044** (0.020)	-0.011 (0.022)	0.046** (0.022)	0.014 (0.018)	0.008 (0.022)
Leadership: age 50+ per 100 students	0.037 (0.025)	0.025 (0.023)	0.004 (0.022)	0.042* (0.023)	0.041* (0.024)	0.023 (0.017)	-0.008 (0.022)
<i>Qualifications reported in the staff directory (all staff)</i>							
Teachers: university qualification share	0.084*** (0.025)	0.099*** (0.024)	-0.030 (0.025)	-0.008 (0.035)	0.052 (0.033)	0.082*** (0.027)	0.052 (0.032)
Teachers: high-school teaching qualification share	0.185*** (0.031)	0.180*** (0.029)	-0.077** (0.032)	-0.048 (0.040)	-0.023 (0.040)	0.131*** (0.027)	0.022 (0.032)
Counselors: university qualification share	0.013 (0.022)	0.003 (0.019)	-0.045** (0.021)	0.023 (0.023)	0.003 (0.027)	0.024 (0.019)	0.040* (0.023)
Counselors: teaching qualification share	-0.003 (0.021)	-0.042** (0.018)	0.023 (0.019)	0.039 (0.032)	0.031 (0.025)	-0.003 (0.020)	0.025 (0.023)
Leadership: university qualification share	0.020 (0.021)	0.014 (0.020)	-0.015 (0.016)	0.037* (0.020)	0.023 (0.023)	-0.035** (0.015)	-0.017 (0.020)
Leadership: teaching qualification share	-0.023 (0.018)	0.000 (0.018)	0.027 (0.017)	-0.001 (0.035)	0.056** (0.026)	0.045** (0.019)	0.066*** (0.023)
<i>Detailed linked-degree measures (staff born in 1989 or later)</i>							
Teachers: high-premium undergraduate share	0.116* (0.067)	0.032 (0.053)	0.518*** (0.062)	-0.025 (0.066)	0.168** (0.066)	0.020 (0.050)	-0.134** (0.053)
Teachers: any high-premium credential share	-0.029 (0.067)	0.032 (0.052)	0.070 (0.058)	0.006 (0.064)	0.040 (0.064)	-0.002 (0.049)	0.046 (0.051)
Teachers: role-specific qualification share	0.012 (0.022)	0.045** (0.022)	-0.054** (0.023)	-0.181*** (0.033)	-0.078*** (0.028)	0.051** (0.021)	0.031 (0.026)
Teachers: role-specific master's share	0.059** (0.023)	0.050*** (0.019)	0.004 (0.017)	-0.011 (0.020)	0.007 (0.022)	0.019 (0.013)	-0.003 (0.018)
Counselors: high-premium undergraduate share	-0.065 (0.082)	0.038 (0.069)	0.058 (0.062)	0.049 (0.090)	0.039 (0.075)	-0.078 (0.062)	-0.046 (0.064)
Counselors: any high-premium credential share	0.139 (0.108)	0.076 (0.077)	-0.029 (0.063)	0.020 (0.086)	-0.069 (0.076)	0.032 (0.053)	-0.001 (0.059)
Counselors: role-specific qualification share	-0.117 (0.076)	-0.041 (0.060)	0.034 (0.051)	-0.090 (0.067)	-0.020 (0.066)	0.067 (0.052)	0.084 (0.062)
Counselors: role-specific master's share	0.054 (0.066)	0.029 (0.049)	-0.069** (0.027)	-0.019 (0.040)	-0.112*** (0.030)	-0.030 (0.027)	-0.076 (0.059)
Leadership: high-premium undergraduate share	0.083 (0.071)	0.013 (0.064)	0.035 (0.066)	0.101 (0.090)	-0.003 (0.071)	0.096* (0.052)	0.133* (0.071)
Leadership: any high-premium credential share	0.042 (0.075)	0.077 (0.057)	0.011 (0.054)	0.049 (0.082)	0.073 (0.070)	-0.016 (0.055)	-0.029 (0.070)
Leadership: role-specific qualification share	0.054 (0.120)	-0.039 (0.098)	0.100 (0.098)	-0.089 (0.118)	-0.033 (0.094)	0.070 (0.068)	-0.022 (0.106)
Leadership: role-specific master's share	-0.118 (0.122)	-0.017 (0.098)	-0.101 (0.095)	0.056 (0.108)	0.034 (0.094)	-0.077 (0.066)	0.036 (0.100)
R^2 : controls only	0.157	0.293	0.058	0.020	0.046	0.486	0.210
R^2 : adding staff (full model)	0.232	0.360	0.421	0.065	0.110	0.530	0.258
Gain from staff measures (ΔR^2)	0.074	0.067	0.364	0.046	0.065	0.044	0.048
Schools	2,058	2,058	2,058	2,058	2,058	2,058	2,058

Notes: Each column is a school-level OLS regression of the All-sample Empirical-Bayes school value added for the indicated outcome (grade-8 cohorts 2017–2020) on the listed characteristics, estimated on public and private-subsidized schools with at least 100 students in the value-added sample. Both value added and continuous regressors are standardized by their sample standard deviations, so coefficients are in standard deviations of value added per standard deviation of the regressor. Staff measures average 2018–2024 administrative records. Staffing ratios divide mean annual headcounts by the number of grade-8-cohort students assigned to the school. University and teaching qualifications are reported in the staff directory and cover staff of all ages. High-premium and role-specific credentials use linked SIES degree records and are shares among staff born in 1989 or later; the SIES register begins in 2007, when this cohort was at most 18 years old. Schools without younger staff in a role have a missing detailed-credential share and enter through the corresponding missing-value indicator. In the estimation sample, these detailed measures are observed for 2,045 schools for teachers, 214 for counselors, and 314 for leadership. All regressions include the administration-type, school-size-category, and curricular-track indicators reported in Table 6; these controls are not displayed. Geography is not included. The sample excludes fully private schools and schools with fewer than 100 students in the pooled value-added sample. Regressions also include indicators for missing regressors, absent roles and incomplete staff rosters. The gain from staff measures is the full-model R^2 minus the controls-only R^2 . Heteroskedasticity-robust (HC1) standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (1) Math achievement; (2) Verbal achievement; (3) High-premium institution; (4) High-premium field; (5) Log projected income; (6) Admission exam taken; (7) Any application (postulación).

teristics account for. The rows add school size and track, staff characteristics, and peer composition in turn, so the last row corresponds to the reported coefficients. Relative to dependency and region alone, size and track raise the R^2 mainly for achievement and admission-exam taking, from 0.090 to 0.194 for math, from 0.128 to 0.330 for verbal, and from 0.251 to 0.505 for exam taking. Staff characteristics add 4.6 and 3.5 percentage points for math and verbal, and between 0.9 and 1.4 percentage points for the remaining outcomes. Peer composition adds the most for achievement, raising the R^2 from 0.240 to 0.416 for math and from 0.365 to 0.521 for verbal, and 5.8 percentage points for exam taking, but at most 1.8 percentage points for institution, field, and financial-aid-application value added. High-premium-field value added is the least explained dimension, with an R^2 of 0.105, and projected-income value added reaches only 0.234.

These results complement the variance–covariance analysis above. The dimensions of school value added that are least related to achievement value added are also the dimensions least accounted for by observable school inputs. Achievement value added is closely associated with who attends the school, while the school effects on field choice and projected income are largely not explained by staffing, credentials, peers, or size. Because these regressions are descriptive, they do not identify the causal effect of any input, and the counselor estimates in particular should be read as suggestive given the number of outcomes considered.

4.4 Persistence of Effects

Another important question to understand these value added effects is whether they persist over time.

(Report a dropout rate?) Considering dropout rate in Chile is not trivial, one might be worried that the VA effects on higher-educational choices are short-lived.

There is an important difference in a school helping a student access higher education and helping them stay in higher education.

For instance,

4.5 Correlation between Value Added Estimates

Table 9 reports the debiased standard deviations (diagonal) and correlations (lower triangle) of latent school value added across the seven primary outcomes, estimated using the cross-cohort procedure in equations (13)–(15).

The debiased standard deviations are larger than the corresponding Empirical-Bayes standard deviations reported in Table 3 for every outcome where the two are comparable: 0.258 versus 0.222 for math, 0.205 versus 0.150 for verbal, 0.123 versus 0.106 for high-premium institution, and, most notably, 0.110 versus 0.053 for high-premium field. This pattern is expected: Empirical-Bayes shrinkage in equation (10) deliberately compresses each school’s estimate toward the across-school mean, which reduces the apparent spread of the posterior means below the true spread of the underlying school effects, especially

Table 8: School value added and second-academic-year persistence

	Higher-ed enrollment	High-premium field	High-premium institution
Higher-ed enrollment VA	0.244* (0.143)	0.055 (0.057)	-0.039 (0.058)
High-premium field VA	0.031 (0.080)	0.032 (0.032)	0.037 (0.035)
High-premium institution VA	0.127 (0.162)	-0.038 (0.068)	0.323*** (0.107)
Math VA	-0.094 (0.058)	-0.003 (0.024)	0.033 (0.027)
Verbal VA	-0.142** (0.064)	-0.036 (0.026)	0.021 (0.029)
Admission-exam taking VA	-0.011 (0.064)	-0.019 (0.025)	0.007 (0.028)
Projected program income VA	0.450*** (0.134)	0.048 (0.055)	0.138** (0.069)
Fin. aid app. VA	0.093 (0.136)	0.014 (0.054)	0.000 (0.053)

Notes: Columns report persistence outcomes requiring entry into the relevant category and continued participation in second academic year. Rows identify the EB observational school value-added measure used as the endogenous school-quality index. In each cell, attended-school VA is instrumented with first-round offered-school VA, controlling for the DA-probable expected value of the same row VA, cohort, grade-4 math and verbal scores, gender, and age. The sample contains timely school applicants from the 2018–2019 cohorts with nondegenerate assignment risk; admission-exam taking is not a sample restriction. Heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

for outcomes like high-premium-field enrollment where individual schools are estimated with less precision.

The correlations reveal three broad patterns. First, the two achievement measures are highly related: math and verbal value added correlate at 0.754. Second, the higher-education-choice outcomes are only moderately related to achievement and, notably, only weakly related to *each other*. High-premium-institution and high-premium-field value added correlate at just 0.145, despite both loading positively on math (0.462 and 0.250, respectively) and on log projected income (0.473 and 0.609). This suggests that “getting students into elite institutions” and “getting students into high-paying fields” are two largely distinct margins along which schools differ, not two symptoms of the same underlying school quality. Third, admission-exam taking and financial-aid application move closely together (correlation 0.597), consistent with both capturing an earlier-stage engagement margin, but both are only weakly related to high-premium-institution and high-premium-field value added (0.075 and 0.060 for admission-exam taking; -0.424 and 0.050 for financial-aid application). The negative correlation between financial-aid-application value added and high-premium-institution value added is the largest-magnitude relationship in the matrix outside the achievement block, and is discussed further below. With the joint-bootstrap standard errors reported in Table 9, the 0.145 correlation between high-premium-institution and high-premium-field value added is precisely estimated and

Table 9: Debiased cross-outcome school value-added correlations

	Math	Verbal	High-inst.	High-pay	Log inc.	Exam	Fin. aid app
Math achievement	0.258 [0.007]						
Verbal achievement	0.754*** [0.027]	0.205 [0.006]					
High-premium institution	0.462*** [0.028]	0.378*** [0.031]	0.123 [0.003]				
High-premium field	0.250*** [0.035]	0.185*** [0.044]	0.145*** [0.039]	0.110 [0.004]			
Log projected income	0.521*** [0.033]	0.432*** [0.038]	0.473*** [0.032]	0.609*** [0.031]	0.106 [0.004]		
Admission exam taken	0.255*** [0.026]	0.328*** [0.030]	0.075*** [0.023]	0.060 [0.040]	0.312*** [0.113]	0.166 [0.003]	
Financial aid application	-0.285*** [0.032]	-0.201*** [0.035]	-0.424*** [0.026]	0.050 [0.034]	0.014 [0.151]	0.597*** [0.020]	0.144 [0.004]

Notes: Debiased school-level value-added standard deviations (diagonal) and correlations (lower triangle), computed with the RSS (Rose, Schellenberg and Shem-Tov) cross-cohort estimator, which removes same-cohort sampling noise from both the variance and covariance terms. Bootstrap SE in brackets below each estimate; for the diagonal this is an independent school-level bootstrap on the variance alone, while for the off-diagonal correlations it is a joint bootstrap that resamples schools once per replicate and recomputes both variances and the covariance together, so the ratio’s sampling variability is captured correctly (500 replicates either way); this is a nonparametric approximation and does not replicate the analytic delta-method standard errors derived in Rose, Schellenberg and Shem-Tov’s Appendix C. Stars on the correlations test the null that the correlation is zero; with 21 pairs tested here, some results significant at conventional levels are expected by chance, and no multiple-testing adjustment is applied. Column headers abbreviate the row labels. Sample: 2017-2020 cohorts, no assignment-risk controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

statistically distinguishable from zero (SE 0.039) despite its small magnitude: the two dimensions are related, just weakly so. By contrast, the correlations of high-premium-field value added with admission-exam taking (0.060, SE 0.040) and with financial-aid-application value added (0.050, SE 0.034) are not statistically distinguishable from zero, the same two relationships flagged as null in the implied-gains discussion below.

Table 10 reports the implied-gains matrix from equation (16), together with joint-bootstrap standard errors.

Because the implied gain is a population-projection coefficient rather than a correlation, it need not be symmetric, and in most cells it is not. A one-unit increase in a school’s true math value added predicts a 0.213 log-point increase in its true log-income value added (SE 0.014); because log-income value added has a much smaller variance than math value added, the reverse projection is larger in magnitude, 1.271 (SE 0.106). Both are precisely estimated and economically large. Most cells in the table are statistically distinguishable from zero at conventional levels, which given 42 directed pairs is itself informative, but three relationships are not: high-premium-field value added is not detectably related to either admission-exam-taking value added (0.074, SE 0.050, and 0.048, SE 0.032, in the two directions) or financial-aid-application value added (0.054, SE 0.036, and 0.046, SE 0.031), and log-income value added is not detectably related to financial-aid-application value added (0.015, SE 0.170, and 0.012, SE 0.135). These null relationships are informative in

Table 10: Implied gains: 1-unit increase in row VA, predicted gain in column VA

	Math	Verbal	High-inst.	High-pay	Log inc.	Exam	Fin. aid app
Math achievement	–	0.597*** (0.023)	0.220*** (0.015)	0.106*** (0.015)	0.213*** (0.014)	0.135*** (0.014)	-0.131*** (0.015)
Verbal achievement	0.953*** (0.051)	–	0.227*** (0.020)	0.099*** (0.024)	0.224*** (0.021)	0.219*** (0.022)	-0.116*** (0.022)
High-premium institution	0.973*** (0.065)	0.629*** (0.052)	–	0.130*** (0.036)	0.407*** (0.028)	0.084*** (0.026)	-0.410*** (0.027)
High-premium field	0.592*** (0.090)	0.343*** (0.083)	0.162*** (0.044)	–	0.586*** (0.042)	0.074 (0.050)	0.054 (0.036)
Log projected income	1.271*** (0.106)	0.836*** (0.090)	0.549*** (0.049)	0.634*** (0.043)	–	0.403*** (0.148)	0.015 (0.170)
Admission exam taken	0.482*** (0.052)	0.493*** (0.046)	0.068*** (0.021)	0.048 (0.032)	0.241*** (0.087)	–	0.519*** (0.022)
Financial aid application	-0.622*** (0.073)	-0.348*** (0.059)	-0.438*** (0.029)	0.046 (0.031)	0.012 (0.135)	0.687*** (0.029)	–

Notes: Row entries give the implied gain in each column outcome’s debiased true school value added from a 1-unit increase in the row outcome’s debiased true school value added, i.e. $\text{Cov}(\text{row}, \text{col})/\text{Var}(\text{row})$, computed with the RSS cross-cohort estimator on the common set of schools used for that pair (matching the correlation matrix’s own denominators). The matrix is not symmetric: the implied gain in column j from a unit increase in row i need not equal the implied gain in row i from a unit increase in column j , since the two outcomes generally have different variances. Standard errors (in parentheses) are from a joint school-level bootstrap that resamples schools once per replicate and recomputes the row and column variances and their covariance together, so the ratio’s sampling variability is captured correctly; this is a nonparametric approximation and does not replicate the analytic delta-method standard errors derived in Rose, Schellenberg and Shem-Tov’s Appendix C. Stars test the null that the implied gain is zero. With 21 outcome pairs (42 directed betas) tested here, some results significant at conventional levels are expected by chance; no multiple-testing adjustment is applied. Column headers abbreviate the row labels. Math and verbal VA are in standardized-test-score units; high-premium-institution, high-premium-field, admission-exam-taken, and financial-aid-application VA are in probability (share-of-students) units; log projected income VA is in log points. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the same way the small correlations above are: they identify specific pairs of outcomes for which a school’s true effect on one carries essentially no information about its true effect on the other, once sampling noise is removed.

4.5.1 IV validation of RSS-implied gains

I next assess whether the gains implied by the RSS covariance matrix predict causal effects in the lottery sample. For each directed pair of outcomes, I multiply the RSS projection of the destination value added on the origin value added by the destination outcome’s same-outcome IV pass-through. This produces a predicted causal gain: the change in the destination outcome expected from a one-unit lottery-induced increase in the origin value added. I then compare this prediction with a direct cross-outcome IV estimate, obtained by instrumenting the origin value added of the attended school with the corresponding value added of the offered school while using the destination outcome as the dependent variable.

This exercise does not constitute a formal test that the RSS-implied and direct IV gains are equal. Such a test requires joint inference that accounts for estimation error and dependence across the RSS covariance estimates, the same-outcome pass-through

estimates, and the cross-outcome IV estimates. The comparison instead provides evidence on whether the RSS-implied gains have empirical content outside the observational sample. To my knowledge, this is the first application to confront gains implied by a cross-outcome value-added covariance matrix with independent lottery-based causal estimates.

Table 11 reports the complete set of estimates in levels. Panel A contains the direct cross-outcome IV estimates, while Panel B reports the corresponding RSS-implied causal gains. Reading the two panels together preserves the economically meaningful scale of each comparison and avoids the instability of ratios when the predicted gain is close to zero.

Table 11: Cross-outcome causal gains: direct IV and RSS-implied predictions

Row school VA	Math	Verbal	Exam taking	High-premium inst.	High-premium field	Log income	Fin. aid app.
<i>Panel A. Direct unshrunk cross-outcome IV</i>							
Math	1.143***	0.694***	0.115***	0.080**	-0.171***	-0.117**	-0.016
Verbal	0.926***	0.947***	0.168***	0.062*	-0.253***	-0.198***	-0.027
Exam taking	0.419***	0.253**	0.470***	-0.050	-0.216***	-0.230***	0.070
High-premium inst.	0.151	0.354	0.024	0.681***	0.371**	0.416***	0.034
High-premium field	0.037	0.246**	-0.079	0.164***	0.708***	0.394***	0.015
Log income	0.458**	0.687***	-0.027	0.362***	0.566***	0.841***	0.153*
Fin. aid app.	0.362	-0.038	0.466***	-0.064	-0.174	-0.102	0.557***
<i>Panel B. Predicted gain: RSS implied gain $\times$ destination-outcome pass-through</i>							
Math	1.143	0.565	0.063	0.150	0.075	0.179	-0.073
Verbal	1.089	0.947	0.103	0.155	0.070	0.188	-0.065
Exam taking	0.551	0.466	0.470	0.046	0.034	0.203	0.289
High-premium inst.	1.112	0.596	0.039	0.681	0.092	0.343	-0.228
High-premium field	0.676	0.325	0.035	0.111	0.708	0.492	0.030
Log income	1.452	0.792	0.190	0.374	0.449	0.841	0.009
Fin. aid app.	-0.710	-0.329	0.323	-0.299	0.033	0.010	0.557

Notes: Panel A instruments attended-school unshrunk value added in each row with first-round offered-school unshrunk value added and changes the student outcome across columns. All regressions control for expected unshrunk row value added, cohort, grade-4 math and verbal scores, gender, and age. Panel B multiplies the RSS latent projection from the row VA to the column VA by the unshrunk same-outcome IV pass-through for the column outcome. Stars in Panel A refer to heteroskedasticity-robust tests against zero. This table provides point-estimate validation; formal tests of equality require joint inference across the RSS and IV estimators. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3 reports all 42 off-diagonal comparisons. The association is positive but imperfect: the correlation between the RSS-implied and direct IV gains is 0.562, and their signs agree in 69 percent of the directed pairs. These patterns provide partial validation of the implied-gain exercise. The RSS estimates contain information about which causal cross-outcome effects are larger and about their direction, but they do not reproduce every IV estimate closely.

Table 12 focuses on ten pairs chosen for their economic interpretation rather than their numerical agreement. A ratio of direct to implied gains equal to one denotes exact agreement. The academic pairs are reasonably close: the direct-to-implied ratios are 1.23 for math value added predicting verbal achievement and 0.85 in the reverse direction. The corresponding ratios are 1.21 when high-premium-institution value added predicts log income and 0.80 when high-premium-field value added predicts log income. The comparison is less successful for some access margins. In particular, the direct effect of exam-taking value added on financial-aid applications is only about one quarter of the

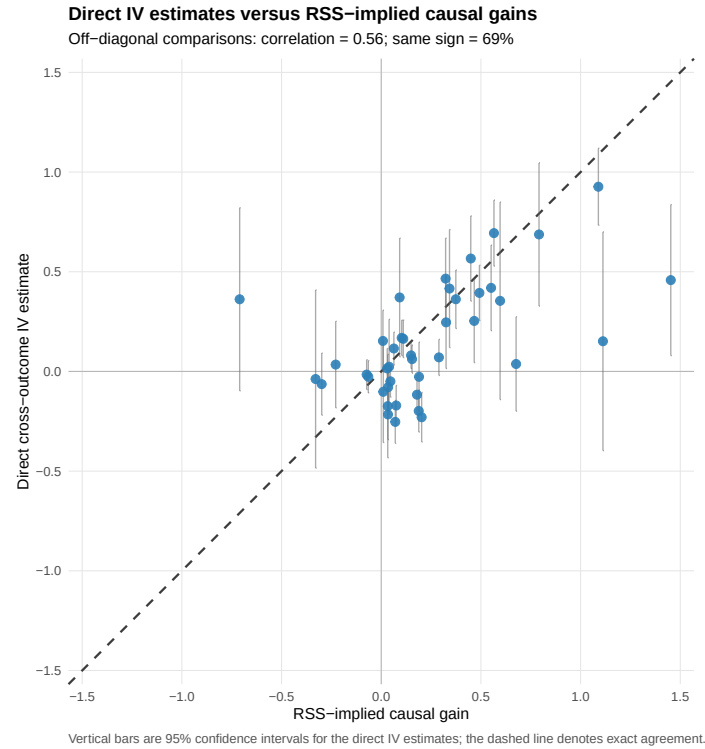


Figure 3: RSS-implied causal gains and direct cross-outcome IV estimates

RSS-implied gain. Thus, the validation is strongest for academic complementarities and the connection between postsecondary quality and projected earnings, while the mapping among earlier application decisions appears less stable.

Table 12: Selected cross-outcome relationships and direct IV validation

Conceptual link	School VA → outcome	RSS projection	RSS-implied gain	Direct IV	Direct / implied
Academic complementarities	Math → Verbal	0.60 (0.02)	0.57	0.69 (0.08)	1.23
Academic complementarities	Verbal → Math	0.95 (0.05)	1.09	0.93 (0.10)	0.85
Academic preparation and access	Math → Exam taking	0.13 (0.01)	0.06	0.12 (0.04)	1.81
Academic preparation and access	Verbal → Exam taking	0.22 (0.02)	0.10	0.17 (0.05)	1.64
Postsecondary complementarities	High-premium inst. → High-premium field	0.13 (0.04)	0.09	0.37 (0.15)	4.02
Postsecondary complementarities	High-premium field → High-premium inst.	0.16 (0.04)	0.11	0.16 (0.05)	1.48
Postsecondary quality and earnings	High-premium inst. → Log income	0.41 (0.03)	0.34	0.42 (0.15)	1.21
Postsecondary quality and earnings	High-premium field → Log income	0.59 (0.04)	0.49	0.39 (0.07)	0.80
Access and financial aid	Exam taking → Fin. aid app.	0.52 (0.02)	0.29	0.07 (0.05)	0.24
Access and financial aid	Fin. aid app. → Exam taking	0.69 (0.03)	0.32	0.47 (0.10)	1.44

Notes: The table reports ten off-diagonal pairs chosen a priori to represent five economically relevant links: academic complementarities, academic preparation and access, postsecondary complementarities, postsecondary quality and earnings, and access and financial aid. The RSS projection maps the row value-added measure into the column value-added measure; bootstrap standard errors are in parentheses. The implied gain multiplies that projection by the destination outcome's unshrunk same-outcome IV pass-through. The direct IV column estimates the cross-outcome effect directly, with regression standard errors in parentheses. A direct-to-implied ratio of one denotes exact agreement. Pair selection does not use the magnitude, sign, or precision of either estimate.

For completeness, Table 13 reports the direct-to-implied ratio for every directed pair. Values near one indicate close agreement, values near zero indicate little direct IV effect relative to the RSS prediction, and negative values indicate opposite signs. The ratio matrix is best interpreted together with the levels in Table 11, since ratios can become very large when the RSS-implied gain is close to zero. Diagonal entries equal one mechanically and therefore provide no validation evidence.

Table 13: Ratio of direct cross-outcome IV gains to RSS-implied causal gains

Row school VA	Math	Verbal	Exam taking	High-premium inst.	High-premium field	Log income	Fin. aid app.
Math	1.00	1.23	1.81	0.54	-2.28	-0.65	0.22
Verbal	0.85	1.00	1.64	0.40	-3.59	-1.05	0.41
Exam taking	0.76	0.54	1.00	-1.07	-6.36	-1.13	0.24
High-premium inst.	0.14	0.59	0.60	1.00	4.02	1.21	-0.15
High-premium field	0.06	0.76	-2.27	1.48	1.00	0.80	0.50
Log income	0.32	0.87	-0.14	0.97	1.26	1.00	17.76
Fin. aid app.	-0.51	0.12	1.44	0.21	-5.32	-9.92	1.00

Notes: Each cell divides the direct unshrunk cross-outcome IV estimate by the corresponding RSS-implied causal prediction, defined as the RSS projection from the row VA to the column VA multiplied by the column outcome’s unshrunk same-outcome IV pass-through. A ratio of one denotes exact agreement, zero denotes no direct IV gain despite a nonzero prediction, and a negative ratio denotes opposite signs. Diagonal entries equal one mechanically because their prediction uses the same same-outcome IV coefficient. Ratios can be large when the predicted gain is close to zero; the underlying levels and standard errors are reported in Table 11.

As a complementary check, I compute how much of each outcome’s raw correlation with math value added survives after removing the debiased linear projection onto math, using the pooled (non-cross-cohort) value-added estimates from equation (2) and the debiased slope $\hat{\beta}^{m|\text{math}}$ from equation (16). For high-premium-institution value added, the naive correlation with math value added is 0.469 before removing the math component and 0.008 after; for high-premium-field value added, 0.232 before and -0.024 after; for log-income value added, 0.425 before and -0.074 after; for admission-exam-taking value added, 0.229 before and -0.026 after; and for financial-aid-application value added, -0.281 before and 0.004 after. In every case, the residual correlation after removing the debiased math projection is close to zero, an order of magnitude smaller than the correlation it started from. This means the debiased implied-gain coefficient captures nearly all of the linear association between math value added and each of the other outcomes: whatever additional signal these outcomes’ value-added measures contain is, by construction, orthogonal to math value added rather than a residual piece of it.

Taken together, these results support the paper’s motivating claim that school effects on higher-education choices are not simply a noisy copy of school effects on achievement. Achievement value added and higher-education-choice value added are related but clearly distinguishable dimensions of school quality, and two higher-education-choice dimensions that might seem similar on their face—getting students into strong institutions and getting students into well-paid fields—are themselves only weakly related. The inference underlying this comparison is currently a nonparametric bootstrap rather than the analytic standard errors in Rose et al. (2022)’s Appendix C, as noted in Section 3; the qualitative pattern of which relationships are precisely estimated and which are indistinguishable

from zero is unlikely to be sensitive to that choice, but the exact standard errors reported here should be treated as a first pass.

5 Conclusion

This paper asks whether high schools causally affect higher-education choices, and to what extent those effects are captured by observational value added. I address this question by combining Chile’s centralized school assignment system with linked administrative data. I first estimate school value added for achievement and higher-education outcomes in a broad national panel, and then use lottery offers to ask whether schools that look better in observational value-added terms also produce larger causal effects.

The main estimates suggest meaningful pass-through. The estimates are 0.884 for math, 0.885 for verbal achievement, 0.497 for high-premium-institution enrollment, 0.803 for high-premium-field enrollment, and 0.311 for log projected program income. In practical terms, schools that observationally look better at raising achievement or higher-education outcomes also predict causal effects on those same outcomes, as I can confidently reject $\psi = 0$ in all dimensions of value added. The results therefore show that the school assignment margin shifts later higher-education choices, not only test scores.

The heterogeneity results show that the relationship is mostly stable across students, with noticeable exceptions. Relative to students in the middle quintile of baseline math achievement, math value added is substantially less predictive in the bottom quintile and marginally more predictive in the top quintile. At the same time, both verbal and math achievement show noticeable gender differences in their forecast coefficients. These differences may suggest some caution when considering achievement value-added disclosure policies, especially in terms of their efficacy and equity across groups. However, the non-achievement results are remarkably stable, making the common-effects approximation more plausible for those dimensions.

Finally, I ask which dimensions of school value added predict gains in projected program income. The ordered horse-race exercise enters math value added first, then high-premium-institution value added net of math, and finally high-premium-field value added net of the first two dimensions. In this joint IV specification, math value added is small and statistically insignificant, while the two higher-education choice dimensions are positive and significant. These results suggest that high schools differ along multiple dimensions in ways that are not captured by achievement alone. In particular, schools affect higher-education decisions through dimensions not captured by math achievement, such as institution and field choice.

These results show why achievement-based measures of school quality are incomplete. As access to secondary and higher education expands, the consequences of schooling increasingly depend not only on learning and enrollment, but also on the decisions students make thereafter. Accountability systems and school-choice policies centered on achievement may therefore overlook school effects that matter for the allocation of skills across fields, occupations, and sectors.

The results also qualify how higher-education choices should be interpreted. Decisions not to continue to higher education, to avoid particular fields, or to attend lower-quality institutions are often interpreted as revealing stable preferences or privately optimal

responses to expected returns. Yet both expected returns and the options students perceive can be shaped by the information, expectations, academic preparation, and support they receive in high school. A decision may therefore be privately optimal given beliefs and preparation that the high school helped produce, while still being causally affected by the school attended. This possibility is especially important in settings where families have limited information about educational options and where access to counseling and other supports varies across schools.

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6 Appendix

Table 14: Balance by school funding type

	Broad Panel (All HS)	Public Funding HS	Private HS
<i>Student characteristics</i>			
Female	0.492	0.492	0.494
Metropolitan Region (Santiago)	0.387	0.366	0.588
Age	13.4	13.4	13.3
<i>Baseline SIMCE variables</i>			
Math 4th grade (z)	0.036	-0.044	0.762
Verbal 4th grade (z)	0.036	-0.033	0.659
Income decile	5.5	5.1	9.4
Mother education years	11.9	11.5	15.5
% matched all SIMCE vars	82.4%	81.9%	88.3%
<i>Middle school</i>			
GPA (z)	0.032	0.029	0.070
Attendance (z)	0.021	0.023	0.010
% matched all middle-school vars	99.8%	99.8%	100.0%
<i>Outcomes</i>			
Std. math exam	0.000	-0.155	1.085
Std. verbal exam	0.000	-0.125	0.886
Admission exam taker	0.776	0.761	0.932
Higher ed. enrolled	0.672	0.654	0.842
Hi-premium institution	0.128	0.089	0.508
Hi-premium field	0.307	0.293	0.444
<i>N</i>	931,547	843,631	87,391
Percent of broad panel	100.0%	90.6%	9.4%

Notes: Broad Panel comprises cohorts 2017–2020 with an observed attended high-school RBD and age 12–16. Public Funding HS includes municipal, local public education service, and subsidized private high schools; Private HS denotes particular pagado high schools. Students with missing school dependency are excluded from both school-type columns. Percentages are relative to the Broad Panel. The SIMCE and middle-school blocks report group means among matched students together with complete-variable match rates. Outcome means use all available observations.

Table 15: Eventual school by SAE application rank and first-round offer status:
VA 2017–2020; SAE 2018–2020

Attendance to target school (column), conditional on (row)	Any of top 3	Choice 1	Choice 2	Choice 3
Receiving offer	69.5% (N = 140,632)	39.3% (N = 140,632)	20.6% (N = 139,954)	12.7% (N = 116,669)
Not receiving offer	33.7% (N = 22,001)	20.1% (N = 22,001)	8.7% (N = 21,778)	6.6% (N = 16,961)
Difference	+35.8 pp	+19.2 pp	+11.9 pp	+6.1 pp

Notes: Each cell reports the share whose most-time high school after grade 8 equals the indicated position in the student's SAE application list; the relevant denominator is shown in parentheses. Rank-specific denominators include students who submitted a school at that rank. Received offer means a positive school identifier in the regular first-round SAE assignment record; it does not include later or supplementary assignment rounds. The difference row reports received offer minus did not receive offer.